Acquisition-Plane Conditioning for Joint Brain Tumor Classification and Segmentation on 2-D MRI: A Controlled Multi-Seed Evaluation With Data-Leakage Audit and External Validation

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***Abstract*—Public 2-D brain MRI benchmarks mix axial, sagittal and coronal slices. We tested whether conditioning a joint tumor classification and segmentation network on the acquisition plane improves performance, using a pre-specified analysis plan. A ResNet-34 encoder feeds a four-class classifier and a U-Net decoder; in the plane-aware variant a learned plane embedding conditions both heads. An otherwise identical model without the embedding served as control. Both were trained with five seeds on BRISC2025 (6,000 T1-weighted slices), on the official split and on a leakage-free split from which 425 development images that duplicated test images had been removed. External evaluation used PMRAM (1,410 images) and AJBDS-2023 (17 patients, 4,826 slices), including patient-wise fine-tuning. Supervising healthy slices with empty masks was essential (false masks on 134/140 healthy test slices without it, 0/140 with it). Plane conditioning had no effect: its seed-level effect on tumor-slice Dice was -0.20 percentage points (95% CI -0.63 to 0.23), equivalent to zero within ± 1 point ($p = 0.003$), and smaller than the 0.57-point difference between two identically configured runs. Ten percent of official test images were near-identical to training images; retraining without them lowered accuracy from 99.6% to 98.1–98.3% and tumor Dice by 1.2–1.3 points. Externally, ensembles classified 94.0–94.8% of PMRAM images correctly, whereas AJBDS-2023 tumor Dice was 36.5–38.0%; threshold calibration and fine-tuning on up to 14 patients reduced false positives but did not improve lesion delineation. Matched multi-seed comparisons, leakage-free splits and external validation are needed before attributing gains to architecture on this benchmark.**

***Index Terms*—Brain tumor, data leakage, domain shift, external validation, magnetic resonance imaging, multi-task learning, segmentation.**

I. Introduction

BRAIN tumor assessment on magnetic resonance imaging (MRI) combines two tasks: deciding which type of lesion is present and delineating its extent. Most deep-learning studies on public 2-D benchmarks treat these tasks separately, training classifiers [1]–[3] or segmentation networks [2], [4] in isolation. Multi-task learning (MTL) with a shared encoder is a natural alternative [5], [6], and joint classification–segmentation models have been reported for glioma grading [7], pituitary adenoma invasion [8], and brain tumor analysis more broadly [9].

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Code and per-image predictions: **[repository URL]**.

Public 2-D brain MRI collections mix slices acquired in three orthogonal planes. The same lesion appears very differently in axial, sagittal, and coronal views, yet standard networks are plane-agnostic. Conditioning a network on acquisition metadata through feature-wise linear modulation (FiLM) [10] is a lightweight way to inject such information and has been used for plane-aware spine segmentation [11]. Whether plane conditioning measurably helps joint tumor classification and segmentation has, to our knowledge, not been tested under controlled conditions.

Answering this question is harder than it appears. On near-saturated benchmarks the differences between design variants are of the same order as run-to-run training variability [12]; comparing an ensemble of one variant with a single model of another conflates ensembling [13] with the design change; and slice-level datasets without patient identifiers are vulnerable to near-duplicate images shared between training and test splits. Reporting guidelines for medical image analysis therefore emphasise matched comparisons, appropriate metrics, and external validation [14], [15].

We study a plane-conditioned joint classification–segmentation network (PAUMT-Net) on the BRISC2025 benchmark [2] with these pitfalls in mind. Our contributions are:

1) A joint network in which a learned plane embedding conditions both the classification head and, through FiLM, a U-Net decoder, together with plane-marginalised inference for data without plane labels.
2) A controlled comparison against an otherwise identical model without plane embedding, using five training seeds per configuration, matched-size seed ensembles, a pre-specified equivalence analysis, and explicit quantification of run-to-run non-determinism and of the effect of ensembling alone.
3) Two data-integrity findings that affect anyone using this benchmark: healthy slices must be supervised with empty masks to avoid false-positive segmentations, and 104 test images (10.4%) of the official BRISC2025 split are near-identical to training images; we quantify their effect by retraining both models on a leakage-free development set.
4) External evaluation on two independent cohorts with pre-specified zero-shot protocols, leave-one-patient-out threshold calibration, patient-grouped fine-tuning with

1 to 14 target patients, and patient-level confidence intervals, including an inference-time slice gate that combines the classification and segmentation outputs.

II. Related Work

Brain tumor classification and segmentation. Convolutional and transformer models reach accuracies close to 99% on curated four-class brain MRI collections [1]–[3], [16]. The BRISC2025 release [2] provides 6,000 T1-weighted slices with expert tumor masks, an official 5,000/1,000 train/test split, and baseline results for both tasks. U-Net variants [4] remain the dominant segmentation architecture; multi-parametric 3-D benchmarks such as BraTS [17] address a different, volumetric setting.

Multi-task learning. Shared encoders can regularise each task [5], [6] but may also suffer from gradient conflicts [18]. Loss weighting by homoscedastic uncertainty [19] and cross-task consistency terms [20] are common coupling mechanisms; we evaluate both as ablations.

Metadata conditioning. FiLM [10] modulates feature maps with an affine transform predicted from side information. Stelzner *et al.* [11] condition lumbar-spine segmentation on the MRI plane. We extend this idea to joint brain tumor classification and segmentation and test it with matched controls.

Evaluation methodology. Metrics Reloaded [14] discusses pitfalls such as empty reference masks and boundary-metric definitions; CLAIM [15] lists reporting items for AI in medical imaging; and benchmark variance across training runs is analysed in [12]. Calibration of modern networks and temperature scaling are described in [21], [22].

III. Materials

A. BRISC2025

BRISC2025 [2] contains 6,000 2-D contrast-enhanced T1-weighted slices in four classes (glioma 1,401; meningioma 1,635; pituitary tumor 1,757; no tumor 1,207) and three planes (axial 1,993; coronal 1,981; sagittal 2,026). We use the official split: 5,000 development and 1,000 test slices (glioma 254, meningioma 306, pituitary 300, no tumor 140; axial 398, coronal 305, sagittal 297). Tumor slices have expert binary masks; healthy slices have none. For each training seed the development set is split 80/20 into training and validation sets, stratified by class $\times$ plane. The dataset provides no patient identifiers, so all internal results are slice-level estimates and patient-level splitting is impossible.

Leakage audit. Because slices from one patient may appear in both official splits, we compared every test image with every development image using MD5 digests and a 64-bit DCT perceptual hash (Hamming distance ≤ 4), and verified each candidate pair by the normalised cross-correlation (NCC) of 128×128 grayscale images. Among 1,000 random test–development pairs the largest NCC was 0.86 (99th percentile 0.66). Seven test images are byte-identical to a development image, and 104 (10.4%; 99 meningioma, 5 pituitary; all with the same label as their match) reach NCC ≥ 0.98 and are called *near-identical*. A further 217 test images have a weaker perceptual-hash match (NCC 0.59–0.97, median above the random-pair maximum), some of which may be neighbouring slices of the same patient. The main results are therefore also reported on the de-duplicated test set ($n = 896$) and on a strict subset that excludes every hash candidate ($n = 679$). The dataset structure and evaluation design are summarised in Fig. 1.

B. External Cohorts

PMRAM [23] contains four-class brain MRI images (glioma, meningioma, pituitary, normal) from three hospitals in Bangladesh, distributed as 512×512 JPEG files. The raw subset of our copy contained 1,505 files; removing exact (MD5) duplicates left 1,410 images, which we use for external classification.

AJBDS-2023 [24] was acquired on a 1.5-T scanner at a hospital in Pakistan and contains slices from several sequences (T1-weighted, T2-weighted, FLAIR, contrast-enhanced T1, ADC, DWI) in all three planes, distributed as 320×320 JPEG images with binary lesion masks. The dataset paper describes annotation of tumor core and edema, so its lesion definition may differ from BRISC2025's tumor masks. Pairing input images with masks yielded 4,826 slices from 17 patients (1,246 with lesion, 3,580 without), which we use for external segmentation. Sequence labels are not provided per slice; most slices therefore differ from the training distribution in contrast as well as in scanner and population.

All three datasets are public and de-identified; this retrospective study used no other data, and no ethics approval was required.

IV. Methods

A. Network

A ResNet-34 encoder [25] (ImageNet initialisation) maps a 224×224 input $\mathbf{x}$ to features $\mathbf{p}_1, \ldots, \mathbf{p}_3$ (64, 128, 256 channels at strides 4, 8, 16) and to $\mathbf{p}_4 \in \mathbb{R}^{256\times7\times7}$ after a 1×1 projection of the last stage. Global average pooling gives $\mathbf{f} \in \mathbb{R}^{256}$.

Plane conditioning (B3). The plane index $p \in \{\text{axial}, \text{sagittal}, \text{coronal}\}$ selects a learned embedding $\mathbf{e}_p = \mathbf{E}[p]$, $\mathbf{E} \in \mathbb{R}^{3\times256}$ (initialised from $\mathcal{N}(0, 0.02^2)$), and $\mathbf{f}_c = \mathbf{f} + \mathbf{e}_p$. The classifier is $\mathrm{LN} \to \mathrm{Linear}(256, 128) \to \mathrm{GELU} \to \mathrm{Dropout}(0.3) \to \mathrm{Linear}(128, 4)$ applied to $\mathbf{f}_c$. The decoder modulates the deepest features with FiLM,

$$\tilde{\mathbf{p}}_4 = \mathbf{p}_4 \odot \sigma(\mathbf{W}_s \mathbf{f}_c) + \mathbf{W}_b \mathbf{f}_c, \tag{1}$$

and three U-Net up-blocks with skip connections from $\mathbf{p}_3, \mathbf{p}_2, \mathbf{p}_1$ followed by bilinear upsampling produce a 224×224 logit map $\hat{\mathbf{M}}$ (Fig. 2). The model has 24.35 M parameters.

Control (B2). B2 is identical except that $\mathbf{e}_p$ is removed ($\mathbf{f}_c = \mathbf{f}$); the FiLM path is kept and conditioned on $\mathbf{f}$ alone. The B2–B3 comparison therefore isolates the 768-parameter plane embedding.

Plane-marginalised inference. External cohorts lack plane labels. For B3 we replace $\mathbf{e}_p$ by the mean embedding $\bar{\mathbf{e}} = \frac{1}{3}\sum_k \mathbf{E}[k]$ (pre-specified) or by $\mathbf{0}$ (sensitivity analysis).

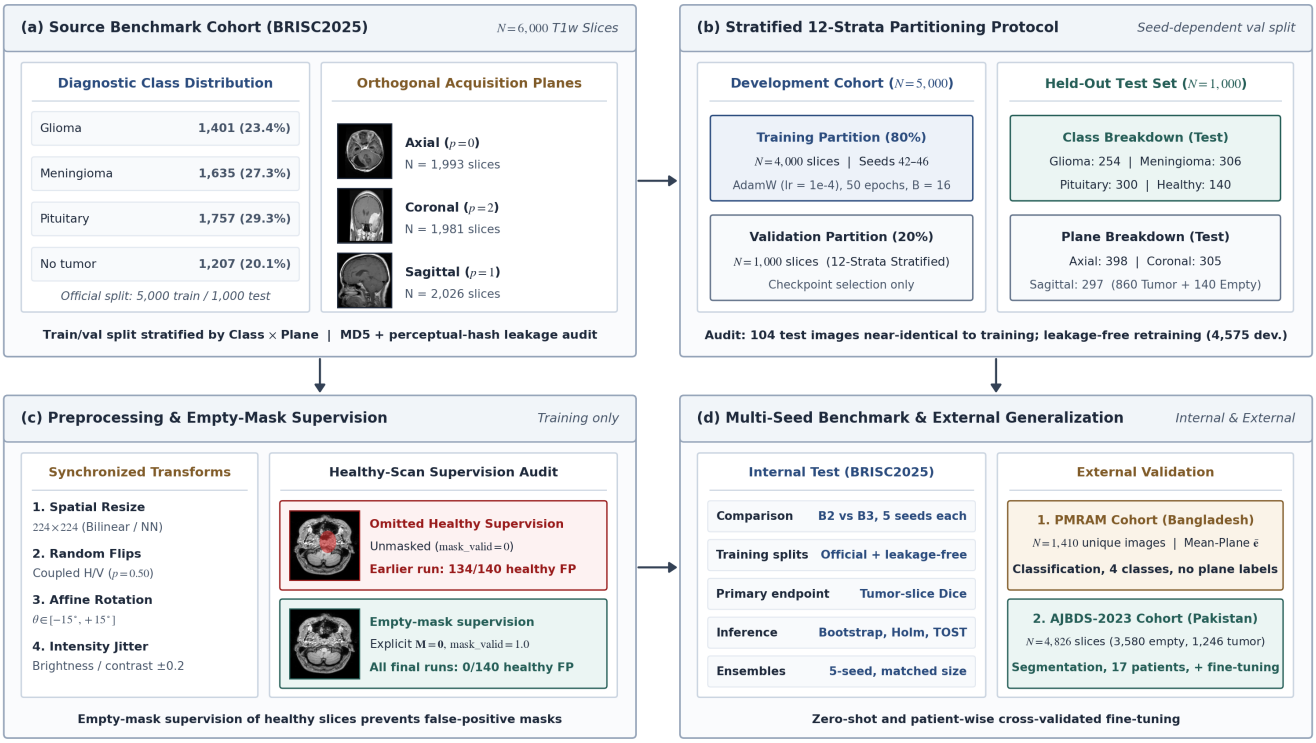


Fig. 1. Data and evaluation design. (a) BRISC2025 class and plane composition. (b) Official train/test split; the validation split is drawn per seed, stratified by class × plane. (c) Augmentation and empty-mask supervision of healthy slices. (d) Internal comparison design and external cohorts.

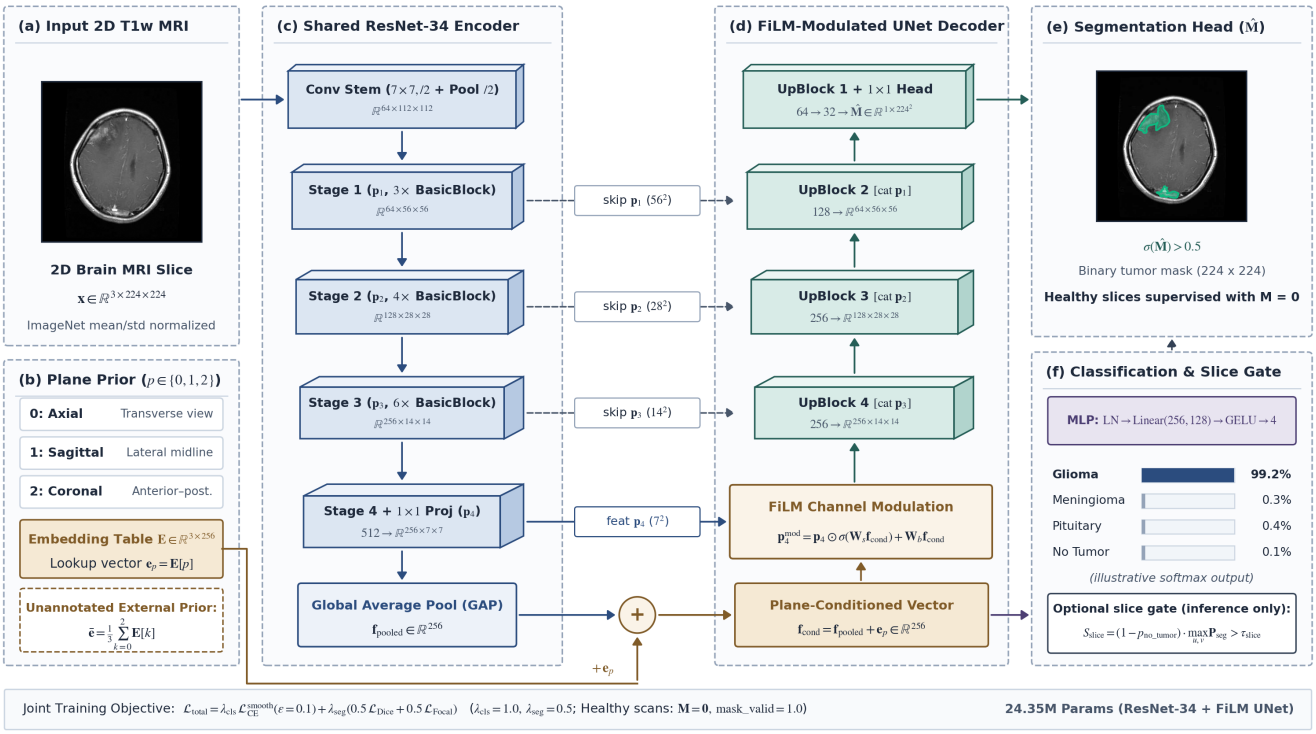


Fig. 2. PAUMT-Net. A shared ResNet-34 encoder feeds a classification head and a FiLM-modulated U-Net decoder; in B3 a learned plane embedding $\mathbf{e}_p$ is added to the pooled feature before both heads. Without plane labels, $\mathbf{e}_p$ is replaced by the mean embedding $\bar{\mathbf{e}}$. The slice gate (bottom right) is an optional inference-time rule evaluated only on external data.

B. Training

The loss is $\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.5\,\mathcal{L}_{\text{seg}}$, with label-smoothed cross-entropy ($\epsilon = 0.1$) for classification and $\mathcal{L}_{\text{seg}} = 0.5\,\mathcal{L}_{\text{Dice}} + 0.5\,\mathcal{L}_{\text{focal}}$ ($\alpha = 0.75$, $\gamma = 2$) [26] for segmentation. *Empty-mask supervision:* healthy slices have no mask file; we assign them an all-zero mask that contributes to $\mathcal{L}_{\text{seg}}$. In an earlier run in which healthy slices were excluded from $\mathcal{L}_{\text{seg}}$, the decoder produced a non-empty mask on 134 of the 140 healthy test slices.

Training has two phases: a 5-epoch warm-up of the decoder alone on the segmentation images of the official training split (which includes the images later used for validation), followed by 50 epochs of joint training with AdamW (learning rate 10^{-4}, weight decay 10^{-4}, batch size 16) and a cosine schedule with 3-epoch linear warm-up. Augmentation consists of random horizontal and vertical flips, rotations in $[-15°, 15°]$ (identical for image and mask; nearest-neighbour for masks), and brightness/contrast jitter (± 0.2). Images are normalised with ImageNet statistics. The checkpoint with the highest validation score $0.5\,\text{F1}_{\text{macro}} + 0.5\,\text{Dice}$ is retained. B2 and B3 were each trained with seeds 42–46; the validation split depends on the seed. Although cuDNN determinism was requested, some GPU kernels are non-deterministic, so runs with the same seed are not bit-identical (Section VI). One run takes about 35 min on a single consumer GPU (NVIDIA RTX 5060, 8 GB).

Ensembles. A seed ensemble averages the softmax probabilities and the pixel-wise sigmoid probabilities of the five models of one configuration before thresholding. B2 and B3 are only ever compared at equal ensemble size.

Calibration. For each single model a temperature T is fitted by minimising the negative log-likelihood (NLL) on that model's own validation split [21]; the temperature-scaled ensemble averages the scaled member probabilities. The test set is never used for fitting.

Slice gate (external data only). Under domain shift a decoder may produce spurious activations on lesion-free slices. We therefore evaluate an inference-time gate that suppresses the predicted mask when

$$S(\mathbf{x}) = \big(1 - p_{\text{no tumor}}(\mathbf{x})\big) \cdot \max_{u,v} P_{\text{seg}}(u, v \mid \mathbf{x}) \le \tau_{\text{gate}}, \quad (2)$$

where both factors are averaged over ensemble members; otherwise pixels are thresholded at 0.5.

C. Leakage-Free Retraining

To measure the effect of train–test overlap directly, we retrained both configurations with seeds 42–46 on a leakage-free development set. It excludes every development image that matched any test image in the audit (MD5 or perceptual-hash candidate, 394 images), and all but one copy of byte-identical development images (38 copies, 7 of them already excluded), leaving 4,575 of 5,000 images. The split, the decoder warm-up and all hyper-parameters are otherwise unchanged, and the official test set is used as is, so it is leakage-free with respect to these models (*LF* models).

D. Ablations

Single runs (seed 42) evaluate classification-only (B0) and segmentation-only (B1) variants and three cross-task couplings on top of B3: concatenating the mean predicted mask probability to the classifier input (A2), adding a consistency loss between tumor probability and predicted mask area (A3, weight 0.1) [20], and uncertainty-based loss weighting [19] (A5). A second B3 run with identical configuration and seed quantifies non-determinism.

V. Evaluation Protocol

A. Metrics

Classification: accuracy, macro F1, macro one-vs-rest AUC, Brier score, NLL, and expected calibration error (ECE, 15 equal-width bins). Segmentation masks are obtained at threshold 0.5. The primary segmentation endpoint is the mean per-slice Dice on the 860 tumor slices. We also report all-slice Dice (healthy slices score 1 when the prediction is empty), which depends on the share of healthy slices and is therefore secondary [14]; tumor-slice IoU, sensitivity and precision (undefined ratios excluded); the number of tumor slices with an empty prediction; and the healthy-slice false-positive (FP) rate (fraction of healthy slices with any predicted pixel). HD95 is the 95th percentile of the pooled symmetric distances between the boundary pixels of prediction and reference [14], computed on tumor slices with a non-empty prediction and reported in pixels at 224×224 resolution; BRISC2025 provides no pixel spacing. Internal Dice uses an additive smoothing constant of 1 pixel; external Dice uses none.

B. Statistical Analysis

Paired differences between models are summarised by the mean difference with a 95% percentile bootstrap confidence interval (2,000 resamples of test slices), complemented by the two-sided Wilcoxon signed-rank test for Dice and the exact McNemar test for accuracy. The Holm procedure [27] controls the family-wise error within pre-defined families, separately for each training setting: (i) B3 vs. B2 at matched size (five seed-matched pairs and the ensemble pair); (ii) each ensemble vs. its own members; (iii) six subgroups (planes, tumor classes) for the ensemble pair. The primary analysis of plane conditioning uses the five seed-level differences in tumor-slice Dice: we report their mean with a 95% t-interval and test equivalence with two one-sided t-tests (TOST) against a margin of ± 1.0 percentage point (pp), which is larger than the difference between two identically configured runs and smaller than gains typically reported on this benchmark; for accuracy the margin is ± 0.5 pp. Equivalence of the ensembles is concluded when the 90% bootstrap CI lies within the margin. Endpoints, margins and the fine-tuning protocol were fixed in a written analysis plan before the additional seeds, the LF models and the fine-tuning experiments were run; one later amendment is described below. Because the Wilcoxon test is sensitive to small consistent shifts in large paired samples, we base conclusions on the effect size and its confidence interval and on the consistency across seeds. For AJBDS-2023, confidence intervals use a patient-level cluster bootstrap.

C. External Protocols

PMRAM is evaluated with the classification head (B3 with $\bar{\mathbf{e}}$; $\mathbf{e} = \mathbf{0}$ as sensitivity analysis). For AJBDS-2023 we pre-specified three protocols: P0, zero-shot without gate (pixel threshold 0.5); P1, zero-shot with gate at $\tau_{\text{gate}} = 0.5$, fixed before inspecting any external result; and P2, target-calibrated, where either the pixel threshold (P2a; grid 0.5–0.95) or τ_{gate} (P2b; grid 0.3–0.95) is chosen by leave-one-patient-out cross-validation: for each patient, the value maximising the balanced score $\frac{1}{2}$(tumor-slice Dice) $+\frac{1}{2}(1 -$ empty-slice FP rate) on the other 16 patients is applied to the held-out patient. P2 uses target labels and is reported as calibration, not as zero-shot performance.

Target-domain fine-tuning (P3). To estimate how much labelled target data restores segmentation, the 17 AJBDS-2023 patients were split into five folds (seeded). For each fold, $k \in \{1, 3, 6, \text{all}\}$ patients of the remaining folds (all: 13–14) were used to fine-tune the seed-42 B2 and B3 models (B3 with the frozen mean embedding) with the segmentation loss; held-out patients were evaluated at threshold 0.5, so every patient is tested exactly once and never seen during its own fine-tuning. The pre-specified protocol used uniform sampling and 600 AdamW steps (learning rate 10^{-4}, batch 16, 50-step warm-up, BRISC2025 augmentation, no early stopping). Because this protocol collapsed to empty predictions even on the training patients (Section VI), we added a post-hoc amended protocol with lesion-balanced sampling (lesion and lesion-free slices drawn with equal probability) and 2,000 steps, chosen solely on the training-patient fit of one fold. Both are reported.

D. Explainability

Saliency maps are computed with LayerCAM [28] for the classification logit of the reference class (B3 seed 42; encoder stage 3 and projected stage 4, averaged), without using the segmentation branch. Localisation is quantified on all correctly classified tumor slices by the fraction of saliency mass inside the reference mask and by the pointing-game hit rate (saliency maximum inside the mask).

VI. Results

A. Internal Performance

Table I summarises the BRISC2025 test results. On the official split, single models classified 99.1–99.7% of slices correctly and reached a tumor-slice Dice of 85.7–86.4%; both 5-seed ensembles reached 99.6% accuracy and 86.3–86.5% tumor Dice. With empty-mask supervision no model produced a mask on any of the 140 healthy test slices, whereas the earlier run without it did so on 134 of 140. HD95 had a median of 2.2 px for every official-split model; its mean (5.7–6.3 px) was driven by a few slices with distant false-positive or missed regions. Per class, ensemble tumor Dice was highest for meningioma (94.0–94.2%), followed by pituitary tumors (87.1–87.4%) and gliomas (75.9–76.4%), whose diffuse margins are also the main source of the lower-decile cases in Fig. 6. Representative median-Dice cases are shown in Fig. 3.

B. Effect of Plane Conditioning

Table II and Fig. 4(A) compare B3 with B2 at matched size. On the official split the five seed-matched differences in tumor-slice Dice were $+0.39$, -0.39, -0.25, -0.27 and -0.48 pp, mean -0.20 ± 0.34 pp (95% CI -0.63 to 0.23); no pair differed significantly after Holm correction. The 90% CI (-0.53 to 0.13) lay within the pre-specified ± 1.0 pp margin, establishing equivalence (TOST $p = 0.003$). Accuracy was equivalent within ± 0.5 pp (-0.08 ± 0.33 pp; TOST $p = 0.023$). The two 5-seed ensembles differed by -0.20 pp tumor Dice (95% CI -0.53 to 0.08) and misclassified the same number of slices (four each).

After leakage-free retraining, the seed-level difference was -0.26 ± 0.77 pp (95% CI -1.22 to 0.70). Equivalence within ± 1.0 pp was only borderline (90% CI -1.00 to 0.47; TOST $p = 0.050$), because single LF models varied more: one seed pair favoured B3 ($+0.92$ pp) and another B2 (-1.15 pp; Holm-adjusted $p = 0.019$ for both). Accuracy differences between LF seed pairs ranged from -1.9 to $+1.1$ pp, so accuracy equivalence could not be shown (TOST $p = 0.21$). The LF ensembles differed by -0.10 pp (-0.39 to 0.19). In subgroups, ensemble differences ranged from -0.58 to $+0.18$ pp on the official split and from -0.25 to $+0.10$ pp after LF retraining, without a direction that was consistent across seeds (Fig. 4(A); Supplementary Table S6).

For scale, two B3 runs with identical configuration and seed differed by -0.57 pp (-1.10 to -0.05), and ensembling alone changed tumor Dice relative to the members by -0.09 to $+1.72$ pp with Holm-adjusted Wilcoxon p values down to 10^{-26} (Table II, bottom).

C. Effect of Train–Test Overlap

Table III compares official-split and LF models on the same test slices. On the full test set the official-split ensembles scored 1.2–1.3 pp higher in tumor Dice (B2: $+1.29$ [0.73, 1.91]; B3: $+1.19$ [0.64, 1.83]) and 1.3–1.5 pp higher in accuracy. The difference was concentrated on the 104 near-identical test images: there, official-split ensembles classified every image correctly and reached 95.5–95.6% tumor Dice, whereas LF ensembles reached 89.4% accuracy and 89.0–89.1% tumor Dice ($+6.4$ to $+6.7$ pp). Most additional LF errors were meningiomas classified as no tumor (9 of 17 errors of the B3 LF ensemble), and 11 of its 17 errors occurred on near-identical images. On the de-duplicated and strict subsets the gap shrank to 0.4–0.6 pp tumor Dice and 0.2–0.6 pp accuracy. Meningioma tumor Dice fell from 94.0–94.2% to 91.3–91.5% with LF retraining. The B3 $-$ B2 comparison was unchanged in every subset (Table III, bottom).

D. Ablations

Table IV lists single-run ablations on the official split. The single-task models were not better than the joint models: the classification-only network (B0) reached 99.10% accuracy, at the lower end of the ten official joint models (99.1–99.7%), and the segmentation-only network (B1) 86.04%

TABLE I
INTERNAL TEST RESULTS ON BRISC2025 ($N = 1{,}000$; 860 TUMOR, 140 HEALTHY SLICES). SINGLE MODELS: MEAN ± SD OVER SEEDS 42–46. ENSEMBLES (5 SEEDS): POINT ESTIMATE [95% BOOTSTRAP CI]. LF: LEAKAGE-FREE RETRAINING.

Model	Acc. (%)	Macro F1 (%)	AUC (%)	Tumor Dice (%)	All-slice Dice (%)	Tumor IoU (%)	Sens. (%)	Prec. (%)	HD95 mean (px)	Healthy FP
Official split										
B2, single	99.40 ± 0.21	99.47 ± 0.18	99.89 ± 0.06	86.06 ± 0.22	88.02 ± 0.19	78.63 ± 0.25	87.58 ± 0.13	87.59 ± 0.33	6.10 ± 0.17	0/140
B3, single	99.32 ± 0.13	99.39 ± 0.14	99.95 ± 0.03	85.87 ± 0.26	87.84 ± 0.22	78.41 ± 0.26	87.79 ± 0.45	87.17 ± 0.33	6.14 ± 0.12	0/140
B2, ensemble	99.60 [99.20, 99.90]	99.65	99.94	86.51 [85.28, 87.58]	88.40	79.22	87.83	88.22	5.73	0/140
B3, ensemble	99.60 [99.20, 99.90]	99.65	99.95	86.31 [85.05, 87.43]	88.22	79.01	88.08	87.68	5.85	0/140
Leakage-free retraining (LF)										
B2 LF, single	97.72 ± 0.72	97.60 ± 0.78	99.66 ± 0.19	84.62 ± 0.69	86.77 ± 0.59	77.07 ± 0.73	86.15 ± 0.26	87.38 ± 0.85	6.56 ± 0.31	0/140
B3 LF, single	97.72 ± 0.56	97.58 ± 0.65	99.65 ± 0.24	84.35 ± 0.16	86.54 ± 0.13	76.77 ± 0.14	86.05 ± 0.32	87.02 ± 0.31	6.60 ± 0.15	0/140
B2 LF, ensemble	98.10 [97.20, 98.90]	97.95	99.89	85.22 [83.86, 86.47]	87.29	77.81	86.43	88.17	6.03	0/140
B3 LF, ensemble	98.30 [97.40, 99.00]	98.15	99.85	85.12 [83.74, 86.38]	87.20	77.67	86.58	87.92	5.90	0/140

Sensitivity, precision and HD95 are computed on tumor slices; HD95 excludes tumor slices with an empty prediction (4–8 per official-split model, 9–18 per LF model). Its median is 2.2–2.8 px for every model. Results for every individual model are given in Supplementary Table S2.

TABLE II
PAIRED COMPARISONS ON TUMOR SLICES ($n = 860$). Δ: MEAN DIFFERENCE IN PERCENTAGE POINTS [95% CI]; SEED LEVEL: t-INTERVAL OVER 5 SEED-MATCHED PAIRS; ENSEMBLES: PAIRED BOOTSTRAP.

Comparison	Δ tumor Dice	TOST p (±1 pp)	Δ acc.
Plane conditioning (B3 − B2), official split			
Seed level ($n = 5$)	−0.20 [−0.63, 0.23]	0.003	−0.08 [−0.49, 0.33]
5-seed ensembles	−0.20 [−0.53, 0.08]	—	0.0
Plane conditioning (B3 − B2), leakage-free retraining			
Seed level ($n = 5$)	−0.26 [−1.22, 0.70]	0.050	0.00 [−1.55, 1.55]
5-seed ensembles	−0.10 [−0.39, 0.19]	—	+0.2
Run-to-run variability (same configuration and seed)			
B3 run 2 − run 1	−0.57 [−1.10, −0.05]	—	+0.2
Ensembling alone (5-seed ensemble − member), range over members			
Official, B2 / B3	+0.16–0.63 / +0.03–0.65	—	−0.1–0.5
LF, B2 / B3	−0.09–1.72 / +0.58–0.96	—	−0.8–1.3

Equivalence margins were fixed before the additional runs: ±1.0 pp tumor Dice, ±0.5 pp accuracy (accuracy TOST p: official 0.023, LF 0.21). Per-pair results with Holm-adjusted Wilcoxon and McNemar tests are released with the code.

TABLE III
OFFICIAL-SPLIT VS. LEAKAGE-FREE (LF) 5-SEED ENSEMBLES ON TEST SUBSETS

Test subset (n)	Model	Acc. (%)	Tumor Dice (%)	Mening. Dice (%)
Full (1,000)	B2	99.60	86.51	93.97
	B3	99.60	86.31	94.16
	B2 LF	98.10	85.22	91.52
	B3 LF	98.30	85.12	91.29
Near-identical only (104)	B2	100.0	95.52	95.60
	B3	100.0	95.64	95.72
	B2 LF	89.42	89.14	88.91
	B3 LF	89.42	88.95	88.70
De-duplicated (896)	B2	99.55	85.27	93.20
	B3	99.55	85.02	93.41
	B2 LF	99.11	84.68	92.76
	B3 LF	99.33	84.59	92.52
Strict (679)	B2	99.41	84.34	92.55
	B3	99.41	84.04	92.78
	B2 LF	98.82	83.77	91.95
	B3 LF	99.12	83.63	91.67

B3 − B2 tumor Dice (pp): official −0.20 (full), −0.24 (de-dup.), −0.30 (strict); LF −0.10, −0.09, −0.14; all 95% CIs include 0.

tumor Dice, within their range (85.7–86.4%). Segmentation-guided classification (A2) and the consistency and uncertainty-weighting variants (A3, A5) lowered accuracy to 98.8–99.1%, and A3/A5 produced one false-positive healthy mask each. Because the identical B3 configuration varied by 0.57 pp tumor Dice between two runs, none of these single-run differences in Dice can be attributed to the design changes; the accuracy losses of A2–A5 (2–7 additional errors relative to the two B3 runs) are likewise within the range expected from run-to-run variation at this sample size.

TABLE IV
SINGLE-RUN ABLATIONS (SEED 42, OFFICIAL SPLIT)

ID	Variant	Acc. (%)	Tumor Dice (%)	Healthy FP
B0	Classification only	99.10	—	—
B1	Segmentation only	—	86.04	0/140
B2	Joint, no plane	99.40	85.88	0/140
B3	Joint + plane (run 1)	99.30	86.27	0/140
B3	Joint + plane (run 2)	99.50	85.70	0/140
A2	B3 + seg.-guided cls.	99.10	85.79	0/140
A3	A2 + consistency loss	98.80	86.21	1/140
A5	A3 + uncertainty weighting	98.80	86.30	1/140

E. Calibration

With label smoothing, the classifiers were under-confident: confidences clustered near 0.92 while accuracy exceeded 97%, giving an ECE of 0.060–0.080 (Fig. 4(B)). Temperatures fitted on the validation splits were 0.51–0.61 and reduced test ECE to 0.001–0.024 and NLL for every model and ensemble (Supplementary Table S3). Calibration did not differ meaningfully between B2 and B3.

F. Explainability

LayerCAM maps of correctly classified tumor slices placed a median of 4.9% of their mass inside the reference mask, 3.8 times the expected share under a uniform map (mean mask area 1.9% of the image). The saliency maximum fell inside the tumor in 59% of slices overall, 92% for meningiomas, 54% for gliomas, and only 29% for pituitary tumors, for which the maps often highlighted surrounding midline structures (Fig. 5). The classification branch therefore attends to the lesion for large, well-circumscribed tumors but not consistently for small sellar lesions.

G. External Evaluation

Table V and Fig. 7 summarise the external results. On PMRAM, single models reached 92.1–92.4% accuracy and 5-seed ensembles 94.7% (B2) and 94.8% (B3). The ensemble

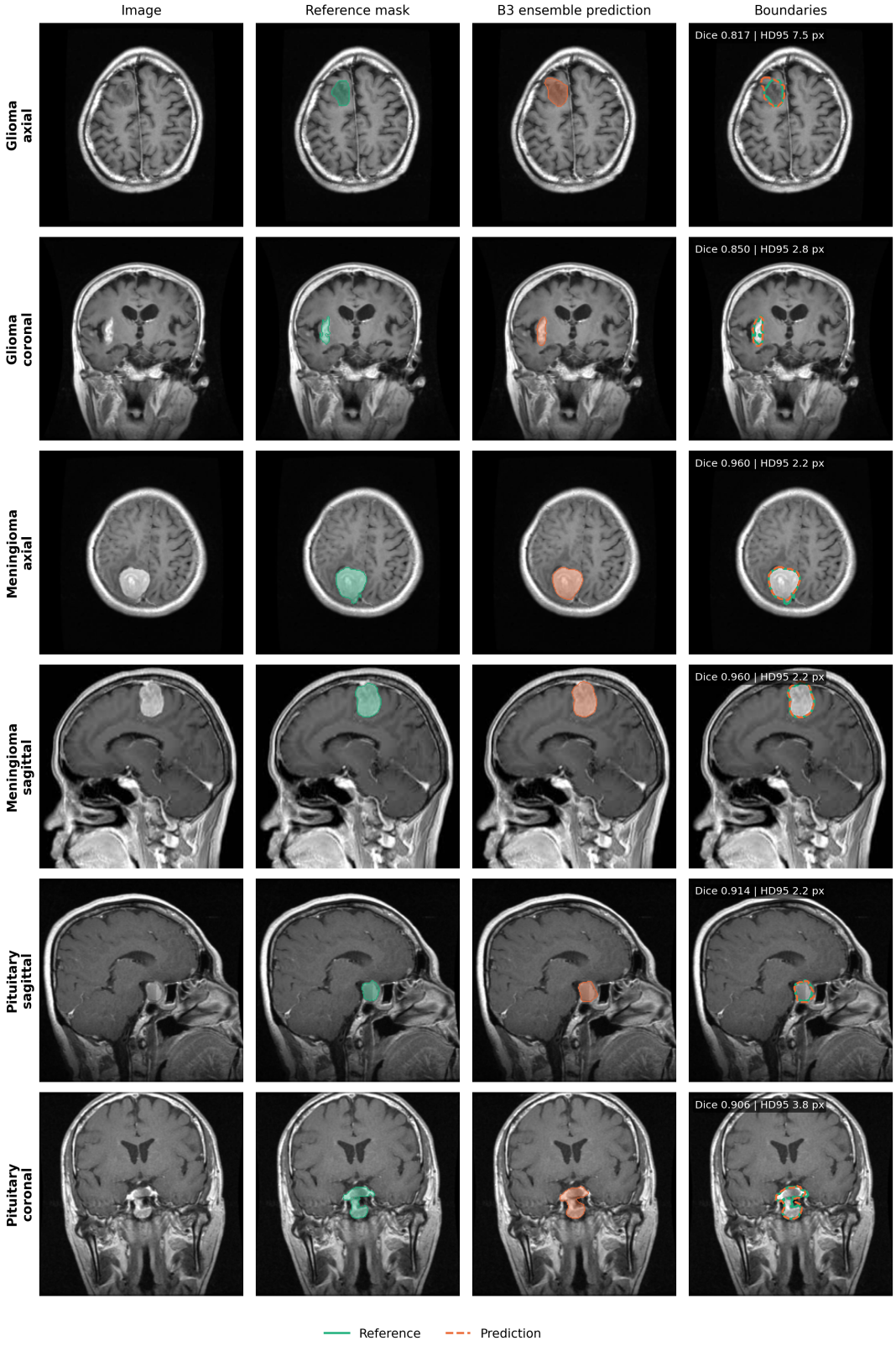


Fig. 3. Representative segmentations of the B3 5-seed ensemble (official split): for each class–plane pair the test slice with the median Dice of that group is shown. Solid: reference; dashed: prediction. HD95 in pixels at 224×224.

difference ($+0.14$ pp; 15 vs. 17 discordant images) was not significant (exact McNemar $p = 0.86$), whereas ensembling improved accuracy over the seed-42 models by 2.5–2.6 pp ($p < 10^{-6}$). Mean-plane and zero-plane inference gave the

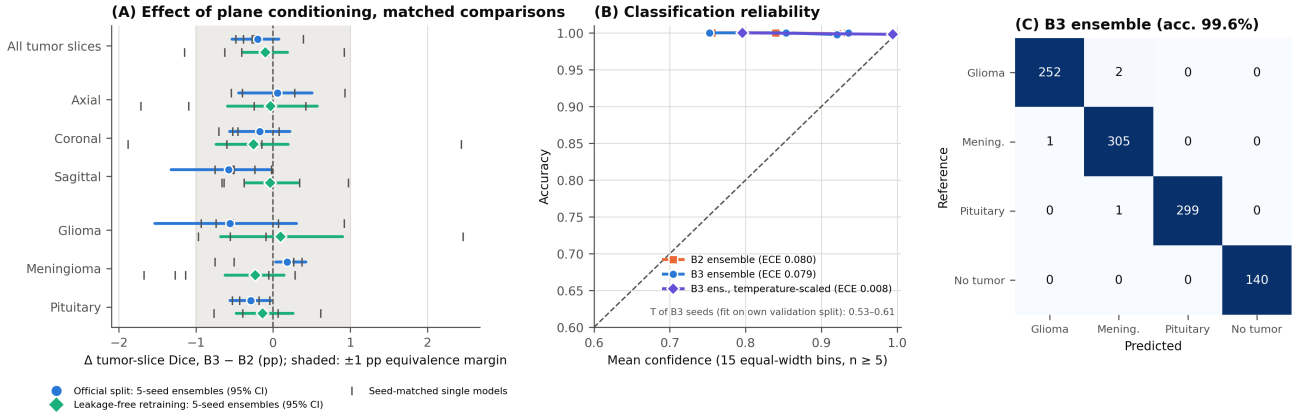


Fig. 4. Internal analysis. (A) B3 − B2 difference in tumor-slice Dice overall and per plane and tumor class: 5-seed ensemble vs. ensemble with 95% bootstrap CI, for the official split (circles) and leakage-free retraining (diamonds); ticks show the five seed-matched single-model pairs; shading marks the ± 1 pp equivalence margin. (B) Reliability diagram (15 equal-width confidence bins with ≥ 5 slices) for the official-split ensembles, and for the B3 ensemble after temperature scaling fitted on each member's validation split. (C) Confusion matrix of the B3 ensemble (official split).

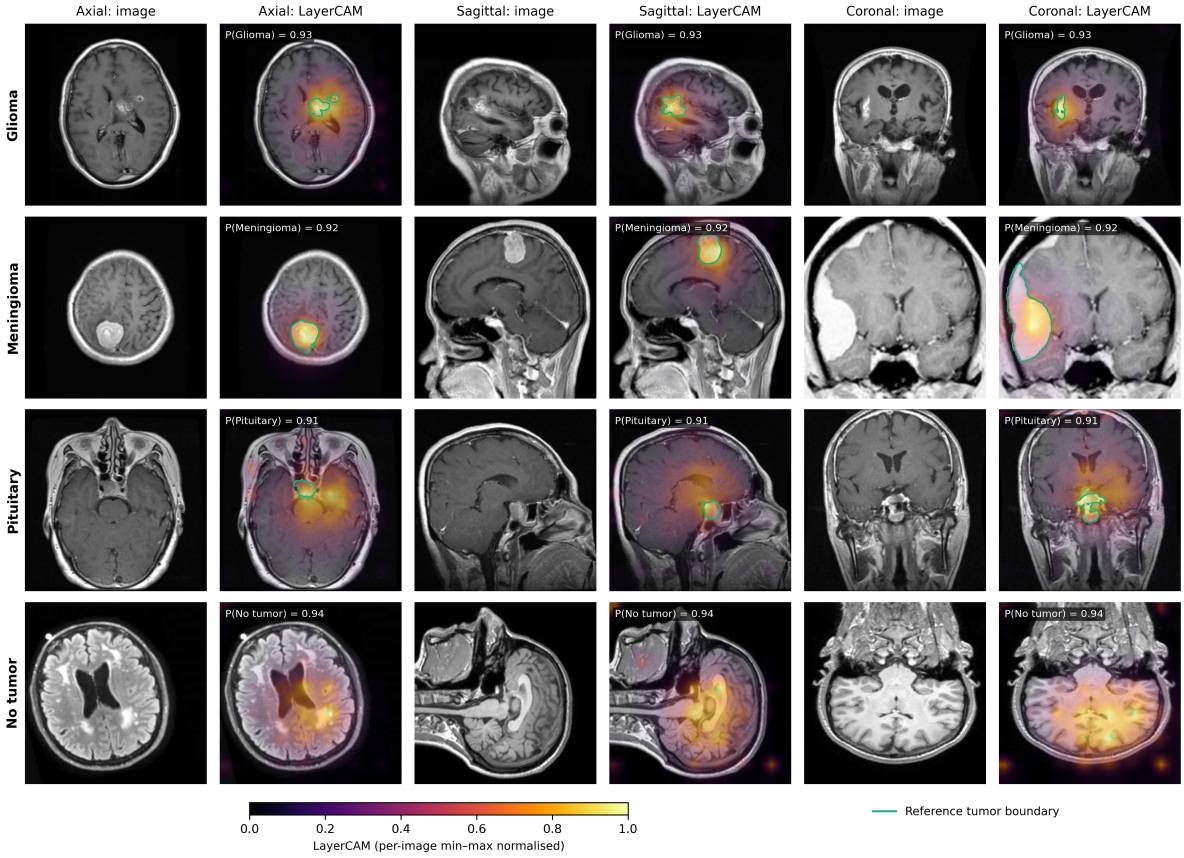


Fig. 5. LayerCAM saliency of the classification logit (B3, seed 42) for one correctly classified test slice per class and plane, chosen as the slice with the median ensemble Dice (tumor classes) or median $P(\text{no tumor})$ (healthy). Maps are min–max normalised per image; the green contour is the reference tumor boundary. Badges show the predicted probability of the reference class, which is close to 0.92 for most slices because of label smoothing.

same accuracy within 0.1 pp. LF ensembles reached 94.0–94.5%, so the internal advantage of the official-split models did not carry over to this external cohort.

AJBDS-2023 exposed a large domain gap. Without any adaptation (P0), ensemble tumor-slice Dice was 36.5% (B2) and 38.0% (B3), 39–41% of lesion slices received no prediction, and 28–31% of lesion-free slices received a false-positive mask; B3 produced more false positives than B2 ($+2.9$ pp [1.6, 4.2]) with a non-significant gain in tumor Dice ($+1.5$ pp $[-0.2, 3.1]$). The pre-specified gate (P1) changed little. Target-

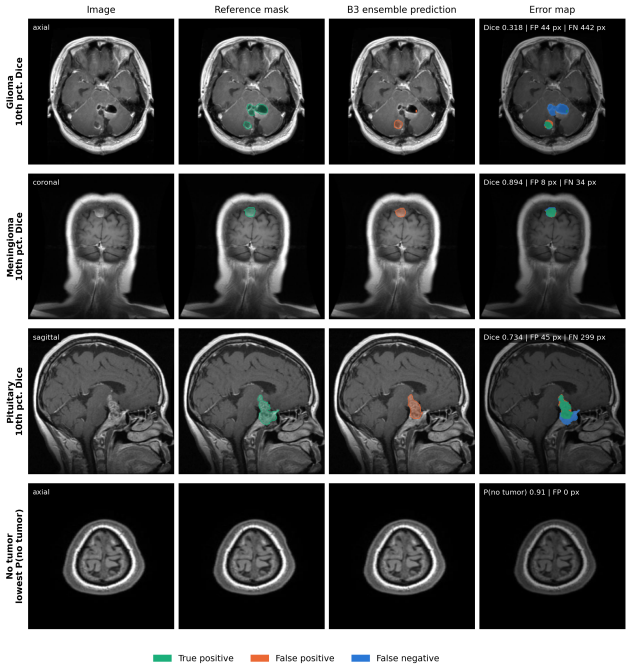


Fig. 6. Lower-decile cases of the B3 5-seed ensemble (official split): for each tumor class the test slice at the 10th percentile of Dice, and the healthy slice with the lowest $P(\text{no tumor})$. Error map: true positive (green), false positive (orange), false negative (blue).

calibrated thresholds (P2) shifted both models towards fewer false positives (12–13%) at the cost of more missed lesions (59–63%); leave-one-patient-out selection chose thresholds of 0.8–0.95. At matched calibration the gate retained more tumor Dice than pixel thresholding at similar false-positive rates (B2: $+3.2$ pp [1.3, 5.3]; B3: $+2.0$ pp [0.5, 3.3]).

Fine-tuning (P3). Under the pre-specified protocol, fine-tuned models predicted almost nothing: held-out tumor Dice was 4.5–15.8% at every k and 72–91% of lesion slices were missed (Supplementary Table S7). The same collapse occurred on the training patients themselves (fold 0, k = all: 8.0% training-patient tumor Dice), identifying an optimisation failure under the 74% share of lesion-free slices rather than a generalisation failure. With the amended, lesion-balanced protocol (Fig. 7(C); Supplementary Table S8), held-out tumor Dice rose with the number of training patients (B3: 5.6%, 17.3%, 24.8% and 26.2% for $k = 1$, 3, 6 and all; B2: 6.7–23.6%) but stayed below zero-shot (B2 36.5%, B3 38.4% for the seed-42 models), whereas empty-slice false positives fell from 32–36% to 5–13% and all-slice Dice rose from 57–60% to 71–77%. Even with all 13–14 training patients, 63–65% of lesion slices in held-out patients were missed, while the fold-0 model reached 55% tumor Dice on its own training patients. B3 and B2 did not differ in held-out tumor Dice at any k; B3 produced fewer false positives at $k \geq 6$ (-7.1 and -7.4 pp).

VII. Discussion

A. Plane Conditioning

Under matched comparisons, explicit plane conditioning did not improve classification, segmentation, calibration, or external transfer. On the official split, its effect on tumor-slice Dice was statistically equivalent to zero within ± 1 pp (mean -0.20 pp over five seed pairs), accuracy was equivalent within ± 0.5 pp, and both effects were smaller than the 0.57 pp difference between two runs of the identical configuration. Leakage-free retraining gave the same point estimate with wider uncertainty. A plausible explanation is that the acquisition plane is trivially recognisable from the image itself, so a 768-parameter embedding adds little information that the encoder does not already extract; marginalising the embedding (mean or zero vector) also left external results unchanged. We therefore report plane conditioning as a tested and negative design choice.

This conclusion differs from the one we would have drawn from a single seed or from comparing an ensemble with a single model. Seed 42 alone suggested a significant gain ($p = 0.033$ before correction), one LF seed pair suggested a gain of 0.9 pp and another a loss of 1.2 pp, and a B3 ensemble outperformed a single B2 model with $p < 10^{-9}$. These impressions disappear under matched comparisons, because ensembling alone produces effects of the same size with very small Wilcoxon p values. On saturated benchmarks such as BRISC2025, where design variants differ by fractions of a percentage point, we recommend multi-seed training, matched ensemble sizes, equivalence testing against a stated margin, and an estimate of run-to-run variability [12].

B. Data Integrity

Two data-related decisions mattered more than the architecture. First, healthy slices without mask files must be supervised with empty masks; otherwise the decoder never learns to predict an empty mask and marks almost every healthy slice. Second, about one tenth of the official BRISC2025 test set is near-identical to training images, almost all of them meningiomas. Retraining without the overlapping images shows the consequence directly: on these images accuracy fell from 100% to 89% and tumor Dice by 6–7 pp, and on the full test set accuracy fell by 1.3–1.5 pp and tumor Dice by 1.2–1.3 pp. Accuracies around 99.5% on the official split therefore partly reflect memorised near-copies; about 98% is a more realistic estimate for unseen slices. Part of the remaining 0.4–0.6 pp gap on de-duplicated subsets may stem from the 8.5% smaller LF training set or from weaker, undetected similarities. Comparative conclusions were unaffected, but results on this benchmark should be reported with a leakage-free training split; we release the exclusion list.

C. Comparison With Published Results

Table VI lists results reported on BRISC2025. Classification accuracies of 98.2–99.7% have been reported, and our official-split models (99.1–99.7%) lie in the same range; leakage-free retraining lowered our single-model accuracy to 97.0–98.9%. The dataset paper reports segmentation as a class-weighted mean IoU (best 80.6%); our per-slice mean tumor IoU (78.4–78.6% seed mean on the official split) is not directly comparable because the aggregation differs. Several studies do

TABLE V
EXTERNAL EVALUATION. PMRAM: 95% BOOTSTRAP CI OVER IMAGES. AJBDS-2023: 95% PATIENT-LEVEL CLUSTER BOOTSTRAP CI (17 PATIENTS). 5-SEED ENSEMBLES UNLESS STATED.

PMRAM, classification (n = 1,410) Model	Acc. (%)	Macro F1 (%)	Macro AUC (%)	ECE
B2, seed 42	92.06 [90.64, 93.48]	92.04	97.31 [96.48, 98.05]	0.037
B3, seed 42 (mean plane)	92.34 [90.92, 93.69]	92.21	97.38 [96.60, 98.14]	0.044
B2 ensemble	94.68 [93.47, 95.82]	94.66	99.25 [98.84, 99.58]	0.061
B3 ensemble (mean plane)	94.82 [93.62, 95.96]	94.81	99.12 [98.68, 99.51]	0.065
B2 LF ensemble	94.47 [93.26, 95.60]	94.45	99.27 [98.92, 99.57]	0.059
B3 LF ensemble (mean plane)	94.04 [92.77, 95.25]	94.01	99.06 [98.64, 99.41]	0.063

AJBDS-2023, segmentation (n = 4,826; 1,246 lesion, 3,580 lesion-free slices) Model	Protocol	Tumor Dice (%)	All-slice Dice (%)	Empty-slice FP (%)	Missed lesion (%)	τ / k
B2 ensemble	P0 zero-shot	36.54 [27.73, 46.10]	63.06 [59.50, 66.40]	27.71 [23.94, 31.67]	41.3	0.5
B3 ensemble	P0 zero-shot	38.00 [29.30, 47.60]	61.26 [58.05, 64.25]	30.64 [26.90, 34.44]	39.3	0.5
B2 ensemble	P1 zero-shot gate	34.88 [25.96, 44.31]	65.26 [61.73, 68.52]	24.16 [20.64, 28.03]	45.9	0.5
B3 ensemble	P1 zero-shot gate	34.81 [26.35, 43.99]	63.57 [60.52, 66.49]	26.42 [23.02, 29.99]	46.7	0.5
B2 ensemble	P2a LOPO pixel thr.	25.94 [17.57, 35.42]	71.12 [67.26, 74.44]	13.16 [10.66, 15.92]	58.8	0.85–0.95
B3 ensemble	P2a LOPO pixel thr.	24.82 [17.11, 34.31]	71.51 [67.70, 74.82]	12.23 [9.67, 15.40]	60.8	0.80–0.90
B2 ensemble	P2b LOPO gate	29.18 [20.34, 38.91]	72.00 [68.30, 75.19]	13.10 [10.42, 15.94]	59.1	0.80
B3 ensemble	P2b LOPO gate	26.82 [18.76, 36.56]	72.38 [68.78, 75.51]	11.76 [9.36, 14.64]	62.5	0.80–0.90
B2, seed 42	P3 fine-tuned†	23.0 [9.3, 37.5]	70.6	12.8 [5.7, 21.0]	64.9	all
B3, seed 42	P3 fine-tuned†	26.2 [14.0, 37.5]	77.0	5.4 [2.0, 10.6]	62.9	all

P0/P1 use no target labels. P2 selects the threshold by leave-one-patient-out (LOPO) cross-validation on AJBDS-2023 labels; τ lists the values selected across folds. †P3: 5-fold patient-grouped fine-tuning with all 13–14 training patients, post-hoc lesion-balanced protocol; held-out patients only. Zero-plane, LF and per-k results are in Supplementary Tables S4, S5, S7 and S8.

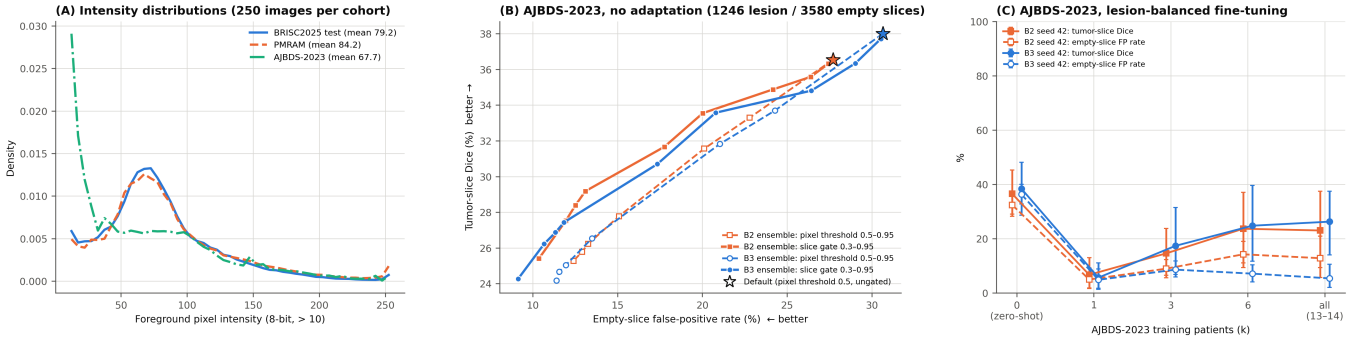


Fig. 7. External domain shift. (A) Distribution of foreground pixel intensities (> 10) in 250 randomly sampled images per cohort. (B) Operating points on AJBDS-2023 for the B2 and B3 5-seed ensembles without target training: pixel-threshold sweep without gate (dashed, open markers) and slice-gate sweep with pixel threshold 0.5 (solid); stars mark the default operating point. (C) Held-out results after lesion-balanced fine-tuning of the seed-42 models with k AJBDS-2023 patients (5-fold patient-grouped cross-validation; $k = 0$: zero-shot); error bars: 95% patient-cluster bootstrap CIs.

TABLE VI
PUBLISHED RESULTS ON BRISC2025 (AS REPORTED BY THE AUTHORS)

Study	Model	Reported result
Fateh *et al.* [2]	ResNet-50	Acc. 98.20 ± 0.80%
	EfficientNet-B0	Acc. 99.20 ± 0.00%
	U-Net / best (SaberNet)	Weighted mIoU 75.7% / 80.6%
Ahmed [3]	ViT, colormap input	Acc. 98.90%, AUC 99.97%
Alkharaan *et al.* [16]	EfficientNet-B1	Acc. 99.70%
This work, official split	B2 / B3, single models	Acc. 99.1–99.7%; tumor IoU 78.2–79.0%
	B2 / B3, 5-seed ensembles	Acc. 99.6 / 99.6%; tumor IoU 79.2 / 79.0%
This work, leakage-free	B2 / B3, single models	Acc. 97.0–98.9%; tumor IoU 75.9–77.7%

Evaluation protocols differ between studies (split, metric aggregation, number of runs); entries are not a ranking.

not state whether the official split was used, none reports multi-seed or leakage-free results, and all are subject to the same train–test overlap, so small differences between methods in Table VI should not be interpreted as rankings.

D. External Generalisation

Classification transferred moderately to PMRAM (92–95% accuracy without retraining), with ensembling contributing about 2.5 pp and plane conditioning nothing. Leakage-free models transferred equally well, consistent with the view that their lower internal accuracy reflects the removal of memorised near-copies rather than weaker features. Segmentation did not transfer to AJBDS-2023: about 40% of lesion slices were missed and 28–31% of lesion-free slices received spurious masks. This cohort differs from BRISC2025 in several respects at once—scanner and population, but also MR sequence (T2-weighted, FLAIR, diffusion and non-contrast T1-weighted slices in addition to contrast-enhanced T1), image export, and possibly lesion definition (edema). The AJBDS-2023 results are therefore best read as a stress test, not as an estimate of scanner shift alone.

Adaptation did not close this gap. Threshold calibration and the slice gate trade missed lesions for fewer false positives along the same curve (Fig. 7(B)); at matched calibration the gate retained 2–3 pp more tumor Dice than pixel thresholding, but no operating point would be acceptable for clinical use. Fine-tuning on up to 14 target patients had the same effect more strongly: it removed most false positives but did not raise held-out tumor Dice above the zero-shot level, even though the models fitted their training patients far better than the held-out ones. With mixed sequences and few patients, lesion appearance apparently varies more between patients than a fine-tuned 2-D model can absorb. The collapse of the pre-specified protocol also shows that segmentation fine-tuning on data dominated by lesion-free slices needs lesion-aware sampling. Gains in all-slice Dice under any adaptation mainly reflect the 74% share of lesion-free slices and should not be read as better delineation. Sequence-aware training or substantially more annotated target data appear necessary.

E. Calibration and Explainability

Label smoothing made all models under-confident on this near-saturated task; a single temperature fitted on validation data removed almost all of the miscalibration. LayerCAM maps localised meningiomas well but pointed to the lesion in fewer than a third of pituitary-tumor slices, which cautions against reading individual saliency maps as evidence that the classifier relies on the lesion.

F. Limitations

BRISC2025 has no patient identifiers, so internal results are slice-level estimates; the leakage audit can only detect near-duplicate images, not all slices from the same patient, and the LF training set is 8.5% smaller than the official one. All models are 2-D and were trained on contrast-enhanced T1-weighted slices only. Five seeds per configuration give moderate precision for seed-level comparisons; equivalence after LF retraining was borderline, and the ablations are single runs whose differences are within run-to-run variability. The decoder warm-up used the segmentation images of the development split, including those later used for validation, which may make validation-based checkpoint selection slightly optimistic; the test set was not involved. HD95 is reported in pixels because pixel spacing is unavailable. The external segmentation cohort comprises only 17 patients; its threshold selection chose values at the upper end of the grid; the fine-tuning experiment used single seed-42 models; and its lesion-balanced protocol was introduced after the pre-specified protocol failed, although it was chosen on training-patient fit only. PMRAM was de-duplicated only for exact copies. Finally, no reader study or prospective evaluation was performed.

VIII. Conclusion

A joint classification–segmentation network reached 99.6% accuracy and 86% tumor-slice Dice on the official BRISC2025 split, with no false-positive masks on healthy slices once these were supervised with empty masks. Conditioning the network on the acquisition plane did not improve any internal or external outcome: across five matched seeds its effect on tumor Dice was equivalent to zero within ± 1 percentage point, and apparent gains in less controlled comparisons were attributable to ensembling and run-to-run variability. Ten percent of the official test set is near-identical to training images; retraining without them lowered accuracy to about 98% and tumor Dice by 1.2–1.3 points, so published results on this split are optimistic. Classification transferred moderately to an external cohort, whereas segmentation failed on a multi-sequence external cohort, and neither threshold calibration nor fine-tuning on up to 14 target patients restored lesion delineation. Future work on this benchmark should use leakage-free splits, matched multi-seed comparisons with equivalence margins, and external validation with substantially more target-domain data.

Data and Code Availability

BRISC2025, PMRAM and AJBDS-2023 are publicly available from their original sources [2], [23], [24]. Training and evaluation code, per-image predictions for all models, the leakage-audit lists, and the scripts that generate every table and figure in this article will be released at **[repository URL]**.

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